Operating Systems 2024/25

Bachelor in Telecommunications and Informatics Engineering
University of Minho

Vítor Francisco Fonte, vff@di.uminho.pt

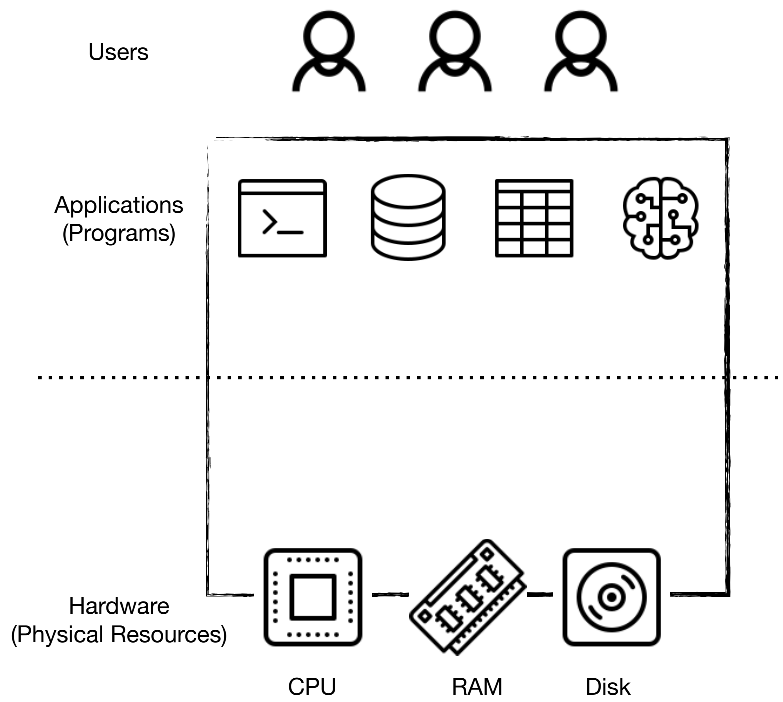
Teaching Team

Name	E-mail	Classes	Office
Vítor Francisco Fonte	vff@di.uminho.pt	T	DI, 2.16
Francisco Nuno Neves	d8328@uminho.pt	PL	DI, 2.14

Note: off-class meetings available but must be scheduled beforehand

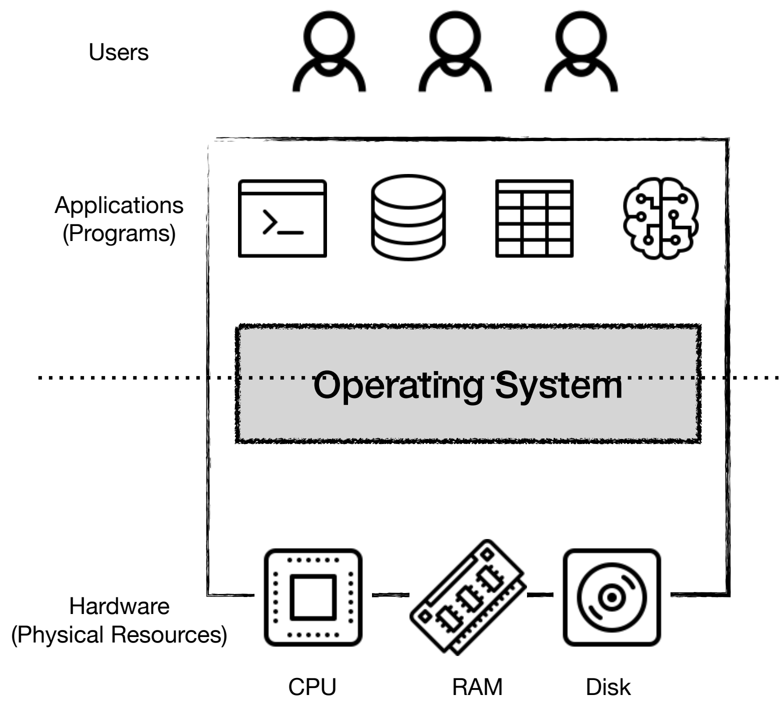
The Landscape

- Applications (Software)
  - Programs executing millions / billions of instructions per second
    (e.g., text editors, databases, AI pipelines)
- Physical resources (Hardware)
  - CPU: where instructions are executed
  - Memory: stores volatile data structures
  - I/O Devices: external devices to interact with
    - Disk: to store data persistently
    - Keyboard, mouse, display, printer, …



The Operating System

- **Mediates** the interaction between programs and hardware
- **Abstracts** physical resources into general, powerful, and easier-to-use virtual forms
- Offers **interfaces** (APIs) to use these resources
  (e.g., run programs, use memory and devices)
- resources for multiple programs
  (e.g., concurrent programs, resource sharing)



Why Bother?

- The OS is a powerful ally to efficiently run multiple programs on a computer, however:
- The incorrect use of OS tools and APIs may lead to
  - Inefficient programs (bad performance)
  - Incorrect programs (bugs, crashes, data corruption)
- To correctly use these tools and APIs one must:
  - Try them! — practical classes
  - Understand them (means knowing the OS) — theoretical classes
- And finally, because Operating Systems can be fun! :-)
  - You will study algorithms, data structures and optimizations applicable to different contexts (AI pipelines, databases, web services, apps, …)

Syllabus
Theoretical classes

- Introduction to modern operating systems
- Process management
- Memory management
- File management
- Device management
- Security management (if time permits)

Syllabus
Laboratory classes

- Will follow a sequence of scripts introducing POSIX APIs related to the topics addressed in theoretical classes
    - Process management
    - File management
    - Pipes
    - Program execution
    - Duplication of file descriptors and I/O redirection
    - Named pipes

Laboratory Classes

- Once in a while, students will be invited to submit solutions to practical challenges
  - The submissions will be graded and feedback will be provided
  - Theses grades will not be used in the student's final grade
- You will need a computer with a basic Linux operating system and a basic C programming language toolchain
  - Can be a virtual machine
  - Installation with textual interface only (no graphical environment) will suffice

Practical Project

- Practical assignment to be developed using the C programming language and the POSIX APIs introduced in the laboratory classes
- Project is mandatory
- Groups of three students
- Setting up of the groups via e-learning
- Submission of assignments via e-learning or GitHub

Final Grade

- 10% continuous assessment
  - Attendance of laboratory classes is mandatory (cf. RAUM)
  - Missing classes may incur in up to 5% penalty in the final mark
- 40-60% individual test (minimum 8 points out of 20)
- 40-60% project (minimum of 10 points, groups of 3 students)
  - Students will be graded individually
  - That means that individual grades may differ within each group

Important Dates

- Written test on Dec 19, 16:00-18:00
- Project due on Dec 10, 23:59
- Project to be presented and discussed on Dec 12+13 (+19?)
  - Penalty for late submission: 10% of the grade obtained for each day
- Optional written test on Nov 21 or 28
  - Opportunity for each student to test his/her knowledge and receive feedback from the teaching team
  - This test will not be used in the final grade

Other Information

- UC activation code in eLearning:
  - SOLETI2024
- E-mail should only be used for:
  - **urgent** scheduling of attendance sessions
  - **urgent** questions relating to the logistics of the course
- **Questions/doubts should be handled in lectures and laboratory classes**

Bibliography

- Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau. Operating Systems: Three Easy Pieces. Arpaci-Dusseau Books, 2018.
  - Freely available at https://pages.cs.wisc.edu/~remzi/OSTEP/
- Avi Silberschatz, Peter Baer Galvin, Greg Gagne. Operating System Concepts (10. ed). John Wiley & Sons, 2018.